Rupankar Dutta

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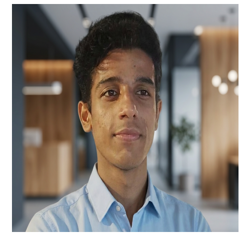

Profile

- B.Tech student at KIIT University specializing in embedded systems, Internet of Things, and edge artificial intelligence, with hands-on experience designing decentralized real-time systems on ESP32. Experienced in developing sensor-integrated firmware, implementing low-latency wireless communication, and deploying lightweight machine learning models for on-device inference. Also proficient in building full-stack applications and system-level solutions combining software, hardware, and data-driven logic.

Education

Kalinga Institute of Industrial Technology (KIIT) 2023–2027
Bachelor of Technology CGPA: 8.4/10

Skills

- **Programming:** Python, C, SQL
- **Embedded Systems / IoT:** ESP32, Sensor Integration, Embedded Firmware, Edge AI, TinyML
- **Machine Learning:** Facial Recognition, MobileNetV2, TensorFlow Lite
- **Backend Development:** FastAPI, Flask, Node.js
- **Web:** React, Socket.IO, MySQL
- **Tools:** Arduino IDE, PlatformIO, EasyEDA, Git, VS Code

Technologies

- **Embedded:** ESP32-WROOM-32, ESP32-CAM, ESP-NOW
- **Sensors:** DHT11, MQ-135, PIR, Sound Sensor
- **ML Stack:** Edge Impulse, TensorFlow Lite
- **Libraries:** BeautifulSoup, PyPDF2

Projects

- **Intelligent Decentralized Home Automation System (ESP32, Edge AI)** – Developed a fully decentralized smart home system using ESP32 nodes communicating via ESP-NOW to support real-time sensing, local decision-making, and offline automation without cloud dependency. Implemented embedded firmware for DHT11, MQ-135, PIR, and sound sensors with timing control, filtering, and threshold-based alert logic, and contributed to the TinyML facial recognition pipeline by building a MobileNetV2-based model for secure on-device access control on ESP32-CAM.
- **Real-Time Multiplayer Quiz Platform** – Designed and built a full-stack multiplayer quiz application using React, Node.js, and Socket.IO for synchronized gameplay across private room-based sessions. Implemented timed rounds, live score tracking, leaderboard generation, and low-latency state synchronization to ensure smooth multi-user interaction and consistent gameplay flow.
- **Natural Language to SQL Query Generator** – Built a Flask and MySQL based system that converts natural language questions into structured SQL queries, making database interaction simpler for non-technical users. Developed the backend flow for query generation, execution, and result display through a clean web interface.
- **Dynamic Programming Text Auto-Correction** – Developed a text correction system in C using edit-distance based dynamic programming to compare user input against dictionary words and generate likely corrections efficiently. Focused on optimized string matching, accuracy, and fast suggestion generation.
- **Automated Plant Watering System (ESP32)** – Built an IoT-based irrigation system using ESP32 and environmental sensing to automate watering decisions based on plant conditions. Integrated real-time monitoring and control features to improve water efficiency and support reliable plant care.
- **WiFi Network Scanner (ESP32)** – Developed an embedded wireless scanning tool using ESP32 to detect nearby WiFi networks and display SSID and RSSI values for quick signal analysis and network diagnostics. Designed it as a compact utility for basic wireless environment monitoring.